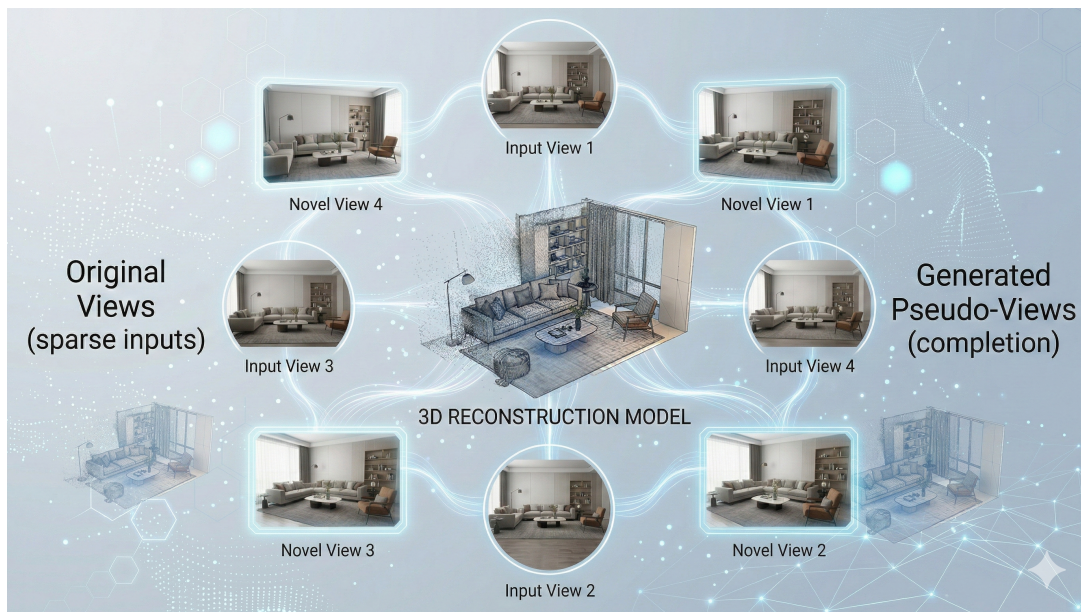


Project 4: Generative Sparse-View 3D Reconstruction

AIAA 3201—Introduction to Computer Vision

Term Project Instruction

Spring 2026



1 Overview

Core Task: The goal of this project is to experience the evolution of 3D reconstruction from “ideal” to “extreme” conditions.

1. Start with **ideal conditions** (dense data) to analyze how different initializations (COLMAP vs. 3D Foundation Models) impact 3DGS convergence and quality.
2. Challenge **extreme conditions** (unknown poses, sparse views) by learning how to register local geometries or perform monocular 3DGS-SLAM without ground truth poses.
3. Explore **Generative AI** to generate missing perspectives, thereby improving the completeness of sparse reconstruction.

Key Concepts:

- **3D Reconstruction:** 3D Gaussian Splatting [8].
- **Geometric Initialization:** SfM (COLMAP) vs. 3D Foundation Models (DUST3R/VGGT).
- **Pose Geometry Estimation:** Registration (Sim3), Optimization, Monocular SLAM.
- **Generative AI:** Diffusion Models for View Synthesis.

Group Size: 2 students per group.

2 Submission Requirements

- **PDF Report (Mandatory):**

- **Format:** Use the CVPR LaTeX Template. The length must be **6–8 pages** (excluding references).
- **ArXiv Upload:** You are required to upload your final report to **arXiv**. Remember to change the “REVIEW version” to “CAMERA-READY version” in template.
- **Content Requirements:**
 1. **Abstract & Introduction:** Background, motivation and solution overview. **Explicitly include your GitHub repository link at the end of abstract.**
 2. **Related Work:** You must review and cite **at least** all papers listed in the Reference Section (Section 4). Expanding beyond this list is encouraged.
 3. **Method:** Explain the technical roadmaps **in your own words**.
 4. **Experiments:** Present quantitative results (Tables) and qualitative analysis (Figures).
 5. **Conclusion:** Summary, limitations, and future work.

- **Code (Mandatory):** Upload your code to a public **GitHub repository** with a clear `README.md` explaining how to run it. Do not submit raw code files. Include dependencies, usage, weights, and visual results in your README.
- **Demo Video (Bonus):** Pack your rendered video results into a `.zip` file and upload it to Canvas. Pack them into a zip file named `videos.zip` and upload it to Canvas. High-quality visualizations will receive **bonus points**.

Important Tips for Success

- **Method Flexibility:** You are encouraged to follow our suggested roadmap, but you are also free to design your own optimal method. Strict adherence to the “Implementation Roadmap” is not mandatory as long as the core tasks are fulfilled.
- **Completeness:** All three parts are **mandatory**. If you fail to complete Part 3, you must discuss the reasons and demonstrate failure cases to receive decent score. Ignoring it completely will result in an “Incomplete” grade.
- **Visuals:** Clean and aesthetic visualizations (e.g., flowcharts, rendering comparisons) are crucial for a Computer Vision course and will significantly improve your score.
- **Dataset Scope:** Completing only the mandatory datasets ensures a passing grade. To achieve a high score, you are expected to report experimental results on additional data.
- **Citations:** Ensure all related works and methods are correctly cited.

3 Guide to Start

To help you get started efficiently, we recommend the following workflow:

1. **Step 1: Literature Review.** Carefully read the papers listed in the **Recommended Reading List** (Section 4). Understanding the core concepts of 3DGS, Sparse Reconstruction, and Generative Inpainting is crucial before coding.
2. **Step 2: Code Familiarization.** Clone the official repositories of the baseline methods (e.g., 3DGS). Run their provided demos to ensure your environment is set up and you understand the input/output formats.
3. **Step 3: Experimentation.** Run the methods on the **Mandatory Datasets** (see Section 6). Adjust code as required by the roadmap. Save your visualization results.
4. **Step 4: Writing & Submission.** Organize your results into tables and figures. Write your report using the CVPR template, citing **ALL** the references. Finally, upload your report to arXiv and submit the PDF report to Canvas.

4 Recommended Reading List

- **3DGS Reconstruction:** 3DGS [8], Scaffold-GS [13], Wavelet-GS [22]
- **3D Foundation Models:** DUST3R [16], VGGT [15], Pi3 [17], MapAnything [7]
- **Unposed Sparse Reconstruction:** InstantSplat [5], S3PO-GS [4], Artdeco [10], RegGS [3], BRPO [21], ReconX [12]

5 Implementation Roadmap

5.1 Part 1: High-Fidelity Reconstruction (Initialization Analysis)

Scenario: Given a video of an object or scene captured with dense views (sufficient data).

Technical Pipeline:

1. Pose & Point Cloud Initialization:

- *Plan A (Traditional):* Run **COLMAP** [1] to extract camera poses and a sparse point cloud.
- *Plan B (Cutting-edge):* Use a **3D Foundation Model** (e.g., VGGT [15], MapAnything [7], or Pi3 [17]) to output zero-shot poses and initial point clouds, then convert them to COLMAP format.

2. 3DGS Optimization:

- Train the scene using the standard **3DGS** [8] pipeline.
- **Key Analysis:** Compare the convergence speed and rendering quality between Plan A and Plan B. Discuss how different initializations (sparse vs. dense, accurate vs. noisy) affect the final reconstruction.
- *Advanced Exploration (Optional):* Interested students are encouraged to replace standard 3DGS with **Scaffold-GS** [13] (voxel-based management) or **Wavelet-GS** [22] (frequency decomposition) to further enhance detail.

Output: Generate novel view renderings for both Plan A and Plan B. Analyze the impact of initialization on the training process and final quality.

5.2 Part 2: Unposed Sparse Reconstruction

Goal: Move away from laboratory settings and handle the challenge of unknown poses and sparse views.

Requirement: Choose **ONE** of the following options (Option A or Option B).

Scenario: Sub-sample the data from Part 1 to create extremely sparse frames (see Section 6), with **NO** camera poses provided.

Option A: Unposed Sparse Reconstruction (Recommended: RegGS [3] or InstantSplat [5])

- **Prediction & Registration:** Use a **Foundation Model** (e.g., DUS3R [16]) or a **Feed-forward Gaussian Model** (e.g., NoPoSplat [20], AnySplat [6]) to predict geometry and align pointmaps/Gaussians into a unified coordinate system.
- **Fusion & Optimization:** Merge the aligned components into a global scene and perform 3DGS post-optimization to refine details.
- *Challenge:* Handling coordinate transformations and scale alignment without ground truth poses.

Option B: Monocular 3DGS-SLAM (Recommended: S3PO-GS [4] or Artdeco [10])

- **System Loop:** Implement a monocular 3DGS-SLAM system to handle continuous video streams.
- **Key Components:** Focus on the interaction between the **Tracking** thread (pose estimation via photometric error) and the **Mapping** thread (Gaussian update with scale-consistent constraints).
- *Challenge:* Preventing scale drift and map collapse in sparse or long-sequence settings.

Output: A 3DGS model reconstructed without COLMAP, along with an evaluation of rendering quality.

5.3 Part 3: Generative Enhancement (Research)

Goal: Use generative models to resolve holes and artifacts in sparse reconstruction.

Scenario: Target the sparse-view scenario defined in Part 2.

Technical Pipeline:

1. **Pseudo-View Generation:** Generate pseudo-views to fill gaps. Choose one approach:
 - Reference the **ReconX** [12] method: Use a **Video Diffusion Model** (e.g., Dynamicraft [19]) to generate views between sparse frames.
 - Reference the **BRPO** [21] method: Introduce the **Difix3D** [18] model to generate pseudo-views by completing rendered images.
 - Implement a view generation method based on other references or your own design.
2. **Hybrid Optimization:** Add the generated pseudo-views to the training set to assist the optimization of **whichever reconstruction framework you utilized in Part 1 or Part 2**.

3. **Confidence Fusion (Advanced):** Since generated views may contain errors, design a simple confidence mechanism (Confidence Mask) to reduce the impact of artifacts on the geometric structure.
4. **Consistency-Aware Optimization (Advanced):** Address temporal flickering and geometric drift in pseudo-views. Use Optical Flow or Reprojection Error to build a strategy that automatically down-weights unreliable regions during training.

Output: Compare rendering results of “Sparse Views Only” vs. “Sparse + Generated Views”. Demonstrate how additional visual information enhances reconstruction.

6 Dataset & Evaluation

6.1 Datasets

- **Mandatory Data:** You can download from Baidu or Google.
 - Waymo-405841 (Outdoor) – P2 Sparsity: 1/10 frames (follow BRPO)
 - D13dv-2 (Outdoor) – P2 Sparsity: 1/30 frames (follow BRPO)
 - Re10k-1 (Indoor) – P2 Sparsity: 1/30 frames
- **Optional Data:** Full Waymo [14]/D13dv [11] datasets mentioned in the S3PO-GS code-base, or Mip-NeRF 360 [2] / Tankand-Temple [9] datasets used in classic 3DGS.

6.2 Metrics

- **Rendering Quality:** PSNR, SSIM, LPIPS (calculated on the Test Set).
- **Pose Error (for Part 2):** ATE (Absolute Trajectory Error) RMSE — Compare the estimated poses estimated against the Ground Truth poses.

6.3 Qualitative Results

You must visualize and compare:

1. **Part 1:** Rendering quality and convergence between [3DGS with COLMAP] vs. [3DGS with Foundation Model Initialization].
2. **Part 2:** Reconstruction results from the Unposed Reconstruction Pipeline (Option A) or Monocular 3DGS-SLAM (Option B).
3. **Part 3:** Reconstruction effects [With vs. Without] Generative View Completion.

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